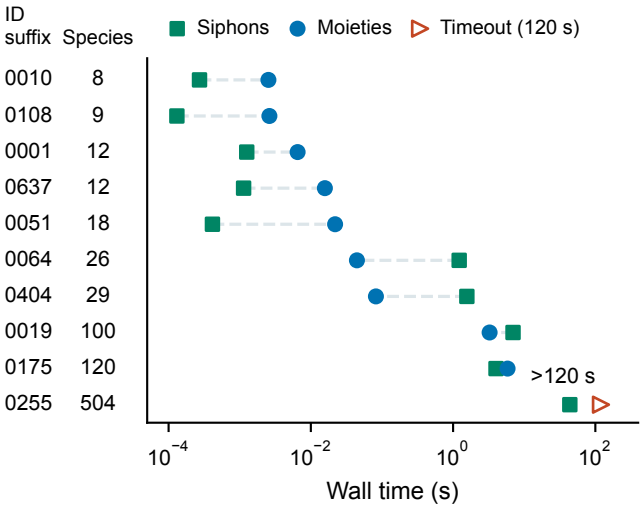
## a Reference checks and independent SBML comparison

| | Species | Rxns | $n$ | $\ell$ | $s$ | $\delta$ | Laws | WR | Tests | External |
|---|---|---|---|---|---|---|---|---|---|---|
| Reversible isomerization | 2 | 2 | 2 | 1 | 1 | 0 | 1 | ● | ✓ | **8/8** |
| Irreversible isomerization | 2 | 1 | 2 | 1 | 1 | 0 | 1 | ○ | ✓ | **8/8** |
| Isomerization cycle | 3 | 3 | 3 | 1 | 2 | 0 | 1 | ● | ✓ | **8/8** |
| Disconnected reversible pairs | 6 | 6 | 6 | 3 | 3 | 0 | 3 | ● | ✓ | **8/8** |
| Pairs with stoichiometry | 6 | 6 | 6 | 3 | 3 | 0 | 3 | ● | ✓ | **8/8** |
| Michaelis–Menten | 4 | 3 | 3 | 1 | 2 | 0 | 2 | ○ | ✓ | **8/8** |
| Reversible Michaelis–Menten | 4 | 4 | 3 | 1 | 2 | 0 | 2 | ● | ✓ | **8/8** |
| Edelstein switch | 3 | 6 | 5 | 2 | 2 | 1 | 1 | ● | ✓ | **8/8** |
| Phosphorylation: one site | 6 | 6 | 6 | 2 | 3 | 1 | 3 | ○ | ✓ | **8/8** |
| Phosphorylation: two sites | 9 | 12 | 10 | 2 | 6 | 2 | 3 | ○ | ✓ | **8/8** |
| Shinar–Feinberg motif | 2 | 2 | 4 | 2 | 1 | 1 | 1 | ○ | ✓ | **8/8** |
| Open reactor | 2 | 3 | 3 | 1 | 2 | 0 | 0 | ● | ✓ | **8/8** |

## b BioModels analysis costs




**10/10 supported structural round trips**

## c Conserved pools in KEGG

**Glycolysis** M00001
14 species · 15 reactions · δ = 1
11 2 1

**Pyruvate oxidation** M00307
13 species · 8 reactions · δ = 1
5 4 3 2 2 2 2 2

**Citrate cycle** M00009
22 species · 12 reactions · δ = 0
8 8 3 3 3 2 2 2 2 2 2 2

**Pentose phosphate** M00004
12 species · 12 reactions · δ = 0
11 11